

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 16: Loci (Locus and its Constructions)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Loci in a triangle

1. Statement 1: The point equidistant from three non-collinear points D , E and F is the circumcentre of $\triangle DEF$. Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true **ICSE 2025, 1 mark**

SOLUTION

A point equidistant from the three vertices lies on the perpendicular bisectors of all three sides. That point is the circumcentre, so statement 1 is true.

The angle bisectors meet at the point equidistant from the three sides, which is the incentre, so statement 2 is true.

Answer: (a)

2. The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes **ICSE 2024, 1 mark**

SOLUTION

The circumcentre is the centre of the circle through the three vertices, so it is the same distance from each vertex.

Answer: (b)

2 Construction questions

3. Using ruler and compasses, construct $\angle ABC = 90^\circ$ with $AB = 6$ cm and $BC = 8$ cm. Construct
 (a) the locus of points equidistant from B and C (b) the locus of points equidistant from A and B
 (c) the locus of points at the fixed distance OA from O , where O is the point common to (a) and
 (b) (d) the locus of points equidistant from BA and BC . **ICSE 2024, 4 marks**

SOLUTION

Draw $BC = 8$ cm, construct a right angle at B and cut off $BA = 6$ cm.

(a) Draw the perpendicular bisector of BC .

(b) Draw the perpendicular bisector of AB .

(c) The two bisectors meet at O . With centre O and radius OA , draw a circle. It passes through A , B and C .

(d) Draw the bisector of $\angle ABC$.

Check: O is the mid-point of the hypotenuse AC , and $AC = \sqrt{6^2 + 8^2} = 10$ cm, so the radius OA should measure 5 cm.

Answer: the loci are two perpendicular bisectors, a circle of radius 5 cm and the bisector of $\angle ABC$

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